

HCI, T1, Introduction To HCI BMIT2203

Question 1

“It is easy to make a *user interface* but it is hard to make a good one” (Lauesen, 2005).

(a) Describe what a user interface is. Give 2 e.g. (*own examples*) to support your answer.

A user interface (UI) is the part of a product, system, or software that the user can see, hear, and touch. It serves as the point of interaction between the user and the system.

Examples:

1. Smartphone Touchscreen – Users interact by tapping icons, typing on a virtual keyboard, and swiping.
2. Voice Assistant (e.g., Amazon Alexa) – Users interact through voice commands and receive audio responses.

(b) Referring to any 2 interactive products/systems/software that you have used before (e.g. *coffee vending machine, GPS, TV remote controller, computer game, e-commerce website, popular word processor, Google Classroom, etc*), write down what it is that you like about each of them and what it is that you do not like. Describe your overall experience in using each of them (e.g. *is it usable? is it fun to use? Does it help you to achieve your goals easily? etc*).

Example 1: Google Classroom

- Like: Simple layout, easy to access assignments and announcements.
- Dislike: Limited customization for organizing classes.
- Experience: Highly usable and effective for managing coursework, but not particularly fun or engaging.

Example 2: Coffee Vending Machine

- Like: Quick and straightforward process for selecting coffee.
- Dislike: Small, hard-to-read labels and occasional payment errors.
- Experience: Functional but frustrating when errors occur; not enjoyable to use.

Question 2

(a) What do you understand by the term “*usability problems*”. Give 2 examples (*own examples*) to support your answer.

Usability problems are issues in a product’s design that make it difficult, inefficient, or frustrating for users to achieve their goals.

Examples:

1. Poorly labeled buttons – Users may not understand what a button does (e.g., “Proceed” vs. “Submit”).
2. Complex navigation – Users get lost in a website or app because menus are unclear or too deep.

(b) Suggest ways to improve the usability of a product/system/software. Explain your answer.

- Conduct usability testing with real users to identify pain points.
- Follow iterative design – build prototypes, test, and refine based on feedback.
- Apply user-centered design – focus on user needs, tasks, and contexts.
- Use clear and consistent labels, icons, and layouts to reduce cognitive load.

Developers should adhere to usability standards and guidelines which are provided by HCI experts such as Nielsen. Following these principles developers can develop user friendly designs and designs that align with users' mental models. These principles include consistency, error prevention, providing clear feedback to user actions and giving control to the users.

Iterative design approach should be used. The iterative design process involves repeated cycles of designing, prototyping, testing, and refining the product. By continuously incorporating feedback from usability testing and user evaluations, developers can identify pain points and enhance the user interface until a good design is achieved.

Question 3

(a) What are the disciplines that contributed to HCI?

The main disciplines contributing to HCI include:

- Computer Science
- Cognitive Psychology
- Ergonomics/Human Factors
- Social and Organizational Psychology
- Sociology

- Design (Interaction, Graphic, Product)
- Anthropology

Computer science - Contributes for the development of the software and hardware of the system and the integration of technology into the system.

Example: Food ordering app

Computer science contributes for the creation of the app's server side logic, handling order process, integration of payment gateways and user authentication. When a user places an order, the backend ensures the order details are sent to the restaurant and payment is processed securely via APIs.

Cognitive psychology - Contributes knowledge of how information is processed in the human mind. It lets us to understand the cognitive capabilities and limitations in terms of memory, attention and perception. By understanding these capabilities and limitations we can design interfaces that align with human cognitive abilities and limitations improving usability and user experiences.

Example: Food ordering app

Cognitive psychology ensures that critical information is easy to remember or retrieve. The app provides a history page where the user can view the past orders eliminating the need to recall, reducing the cognitive load of the users. In addition, the app should provide visual cues like icons or thumbnails to help users recognize dishes faster.

Social and organizational psychology - Contributes knowledge of users' social interactions, cultural influences and workplace behaviours.

Example: Food ordering app

In culturally diverse regions, people might have dietary restrictions or preferences (e.g., vegetarian, halal). The app can include filters for dietary preferences and clearly label items. Notifications about local festivals or special menu items based on user location or cultural habits.

Ergonomics and human factors - Contributes knowledge of designing a system for user comfort, efficiency, and safety. They focus on reducing physical and cognitive strain, enhancing usability and users productivity.

Example: Food ordering app

Ergonomics/human factors should ensure the layout and design of the app are visually and physically accessible. For example, buttons should be sized appropriately for touchscreens to avoid misclicks, especially for users with larger fingers. In addition, high-contrast color schemes and adjustable font sizes make the app usable for visually impaired users.

(b) Dix (2004) stated that, “*It is not possible to design effective interactive systems from one discipline in isolation*”. Explain why this is the case.

Effective interactive systems require a multidisciplinary approach because:

- Computer Science provides technical capabilities and limitations.
- Cognitive Psychology helps understand user memory, attention, and perception.
- Ergonomics ensures physical and environmental fit.
- Social Sciences address cultural, organizational, and contextual factors.

Designing in isolation from any of these can lead to systems that are technically sound but unusable, unsafe, or inappropriate for the target users and contexts.

Additional questions (on your own, i.e. questions below will not be discussed during tutorial) List 3 benefits of system/software/product with good interface design.

List 3 negative effects of system/software/product with bad interface design.

Read the following paper/web page and highlight some points that made the most impact on you. Be focus on issue related to HCI.

(a) “Guaranteeing Rights for the User” by Clare-Marie Karat. <http://www.research.ibm.com/compsci/spotlight/hci/p29-karat.pdf>

(b) “Ubuntu's Mark Shuttleworth on shaking up system software” by By Leo Kelion.

<http://www.bbc.com/news/technology-17916879>

Past year question (Group 4)

You are designing a mobile app for the elderly to report their loss during the recent floods. They only need to provide photos of their house or damaged items, and their personal information to claim the flood aid cash vouchers from the government. Answer the following questions based **on the scenario**.

- a) The study of Human Computer Interaction (HCI) involved multiple disciplines. Explain how **Computer Science** and **Cognitive Psychology** fields contributed to HCI design of this mobile app. (5 + 5 marks)

- Computer Science:
 - Provides the technical foundation to build the app, including the software architecture, database management for storing user information and photos, and the algorithms for image processing and data submission.
 - Ensures the app is functional, secure, and can handle the data reliably, for example, by implementing a simple and secure file upload system for photos and an encrypted connection for submitting personal details.
- Cognitive Psychology:
 - Helps understand the mental capabilities and limitations of elderly users, such as potential declines in memory, attention, and perception.

- Informs the design to reduce cognitive load. For example, using large, clear text, simple icons, a linear step-by-step process for uploading photos and filling forms, and clear confirmation messages to ensure users understand each action and its consequence.

Past year question (Group 5)

You want to build a meal ordering website for TARUMT café. This website is aimed at helping students order their meals before they are sold out. Students can order their desired meals a few hours before their break time and pay via online banking, e-wallet, or debit/credit card. Answer the following questions **based on this scenario**.

- b) Explain with **ONE (1)** example of effect to the students, and **ONE (1)** example of effect to the website developer if this website has a **BAD** usability design. (5 + 5 marks)

- **Effect on Students (Users):**

- Example: If the navigation is complex or the payment process is confusing, students might struggle to complete their orders successfully. This could lead to frustration, wasted time, and ultimately, them missing the chance to order their meal before it sells out. They might abandon the website and revert to less efficient methods.

- **Effect on Website Developer:**

- Example: A website with bad usability will lead to a high volume of customer support calls and emails from confused students. This increases the cost and time the development team must spend on maintenance and user support, rather than on improving the system or developing new features. It can also damage the developer's reputation.

Past year question (Group 1, 2)

You are designing a mobile app for the elderly to report their loss during the recent floods. They only need to provide photos of their house or damaged items, and their personal information to claim the flood aid cash vouchers from the government. Answer the following questions **based on the scenario**.

- b) Explain **ONE (1)** effect to the elderly users and **ONE (1)** effect to the app developer, if your app has a **GOOD** usability design. (5 + 5 marks)

- **Effect on Elderly Users (Users):**

- Example: A good usability design would make the process of reporting losses less stressful and more efficient. With large buttons, clear instructions, and a simple workflow, elderly

users can complete the claim process confidently and successfully without needing assistance, reducing their anxiety during a difficult time.

- Effect on App Developer:
 - Example: A well-designed, usable app will minimize the cost of customer support. Fewer users will encounter problems or get stuck, leading to fewer help-desk calls. This saves the development team significant resources and time, allowing them to focus on other projects or updates. It also builds a positive reputation for the developer.

HCI, T2, User Analysis & Task Analysis BMIT2203

Question 1

“To the user the UI is the system. Most interfaces need to be designed to accommodate both the novice 新手 and expert users at the same time.”

Do you agree with the above statement? Give 2 examples (**own examples**) to support your answer.

Yes, I strongly agree with this statement.

1. The UI is the System: For the vast majority of users, the interface is their only point of contact with the underlying technology. They cannot see or interact with the code or databases; they only experience the functionality through the UI. A poorly designed UI, therefore, makes the entire "system" feel poor, regardless of its technical sophistication.
2. Accommodating Novice and Expert Users: Effective systems are often used by a mix of users. A design that only caters to one group alienates the other.
 - Example 1: A Photo Editing Software (e.g., Adobe Photoshop)
 - For Novices: It features guided edits, simple sliders for adjustments, and clear tooltips.
 - For Experts: It provides complex layer masks, customizable keyboard shortcuts, and advanced scripting capabilities.
 - Example 2: A Library Self-Service Kiosk
 - For Novices (e.g., a first-time user): It has large, clearly labeled buttons ("Borrow," "Return"), on-screen instructions, and a simple, step-by-step process.
 - For Experts (e.g., a librarian or frequent user): It might allow scanning multiple books in quick succession without confirmation prompts for each item, or have a shortcut key to jump directly to the inventory check screen.

Question 2

(a) How do the requirements of an expert user differ from those of a novice? Present your answer in a table format.

Aspect	Novice User Requirements	Expert User Requirements
Guidance	Needs clear, step-by-step instructions and prompts from the system.	Prefers minimal interference; brief or no feedback.
Input	Requires simple, brief inputs; not assumed to be proficient.	Prefers efficient input methods like keyboard shortcuts and command sequences.
Training	Should not require special training; all info must be within the system.	Willing to learn advanced features to optimize workflow.
Options & Complexity	Needs a small set of clear options to avoid feeling overwhelmed.	Demands access to advanced functions, settings, and features.
Feedback & Help	Requires sufficient, clear, and unambiguous feedback and help.	Prefers brief feedback and comprehensive, but non-intrusive, help facilities.
Interaction Initiation	Prefers the system to lead the interaction.	Prefers to initiate actions and control the flow.

(b) If you are developing a system that needs to cater for both the novice and expert users at the same time, which category of user would you pay more attention to? Give reasons for your answer.

I would pay more initial attention to designing for the novice user.

Reasons:

1. **Lower Barrier to Entry:** If a novice user finds the system confusing or difficult to learn, they will likely abandon it entirely. A design that is intuitive for a novice ensures user acquisition and retention. It is easier to add advanced, power-user features onto a simple, solid foundation than it is to simplify a complex system built for experts.
2. **Error Prevention:** Novices are more prone to errors. A design that focuses on their needs (clear labels, confirmation steps, good feedback) inherently makes the system more robust and safer for all users, including experts.
3. **Foundation for Expertise:** The core tasks and navigation paths that are designed to be clear for novices form the fundamental model that experts will also use. Once experts learn this basic model, they can then efficiently utilize the advanced shortcuts and features layered on top.

Question 3

To design an effective interactive system, it is necessary to know the answers to the following 3 questions:

1. Who are the users?
2. What are the tasks?
3. What is the environment in which the system will operate?

Give reasons why it is important for a designer to have the answers to the above 3 questions?

1. Who are the users?

Knowing the users allows the designer to create **personas** and understand their **capabilities, limitations, and motivations**. For example, designing a system for children requires large buttons and simple language, while designing for financial analysts requires support for complex data and rapid input. Without this knowledge, the design may mismatch the users' skills, knowledge, or goals, leading to poor usability.

2. What are the tasks?

The ultimate purpose of a system is to **help users achieve their goals**. Task analysis (like Hierarchical Task Analysis) helps designers understand what users are trying to accomplish and the steps they take. This ensures the system provides the necessary functionality in a logical and efficient workflow. Designing without understanding the tasks can result in a system that is feature-rich but doesn't actually help users complete their work effectively.

3. What is the environment?

The context of use profoundly impacts design decisions. A system used in a bright, sunny outdoor setting needs a high-contrast, glare-resistant screen. A system used in a quiet library should have silent feedback mechanisms. A system used by someone who is frequently interrupted needs to

easily save progress. Ignoring the environment can render a system unusable in its intended context, no matter how well it is otherwise designed.

So that they can identify the users, their background, goals and values, and their mental models. Once users are identified, the designer needs to perform task analysis on the way users perform their current job with an existing system. The designer then identifies these tasks and translates them into a user interface. The environment is where the system provides the physical inputs/outputs. This is so that the UI design used is suitable for the specified environment.

Past Year Question (Group 4)

What is Persona? Discuss why it is an important tool at the analysis stage of Human computer interaction (HCI) framework.

- What is a Persona?

A persona is a fictional, detailed user model representing a major user group for a system. It is not a real person but is synthesized from real-world observations and data gathered during user research. A persona includes details like a name, photo, demographics, goals, motivations, skills, attitudes, and pain points (e.g., "70-year-old Pak Ali, not comfortable with technology, wants a simple way to report flood damage").

- Importance at the Analysis Stage:

1. Provides a Shared Understanding: It ensures all team members (designers, developers, stakeholders) have a consistent and clear understanding of who they are designing for, preventing assumptions.
2. Keeps the Design User-Centered: By giving the user a face and a story, personas help the team empathize with the user's needs, frustrations, and context, ensuring design decisions are made with a specific user in mind rather than based on personal preference.
3. Guides Design Decisions: Personas help answer critical design questions by asking, "What would Pak Ali need here?" or "Would Siti be able to understand this instruction?" This makes the design process more focused and effective.

Past Year Question (Group 5)

Discuss why HTA is an important tool to the user interface designer at the analysis stage of HCI framework.

- **What is HTA?**

Hierarchical Task Analysis (HTA) is a method for breaking down a user's main goal into a hierarchy of subtasks. It involves identifying what the user needs to do (tasks) and the plans for the order in which these tasks are performed.

- **Importance at the Analysis Stage:**

1. **Understands Current User Behavior:** HTA helps the designer understand in detail how users currently perform their tasks, whether using an existing system or a manual process.
2. **Informs Compatible System Design:** By understanding the user's existing task structure, a new system can be designed to be compatible with the user's accustomed workflow. This means skills learned from the old system can be transferred, reducing the learning curve for the new system.
3. **Identifies Key Tasks and Flows:** It helps pinpoint the most critical and frequent tasks, ensuring the UI is structured to support these efficiently. It can also reveal redundant or complex steps that can be simplified in the new design.

Past Year Question (Group 3)

You want to build a **meal ordering website** for TARUMT café. This website is aimed at **helping students order their meals** before they are sold out. Students can order their desired meals a few hours before their **break time and pay via online banking, e-wallet, or debit/credit card**. Answer the following questions based on this scenario.

Explain the **purpose** of using Scenario during the **analysis stage of this app development**.

The purpose of using a scenario during the analysis stage is to create a concrete, narrative description of how a specific persona (e.g., "Ahmad, a busy final-year student") would use the system to achieve a goal in a realistic context.

- **Brings Requirements to Life:** It moves beyond abstract feature lists (e.g., "must have a payment gateway") and illustrates how and why a feature would be used in a real-world situation (e.g., "Ahmad is between classes, using his phone on the campus WiFi to quickly re-order his usual meal so he can pick it up without queuing").
- **Identifies User Needs and Context:** Scenarios help uncover functional and non-functional requirements by considering the user's environment, motivations, and potential problems. For example, the scenario might highlight the need for a function or the importance of the website working well on a mobile device under potentially spotty network conditions.
- **Facilitates Communication:** It provides a simple, story-based tool that all stakeholders (designers, developers, clients) can understand and discuss, ensuring everyone has a shared vision of the user's interaction with the system.

Past Year Question (Group 4, 5)

- a) Explain **ONE (1)** advantage and **ONE (1)** disadvantage of using interview technique compared to questionnaire technique. (5 + 5 marks)

Advantage

Rich, Qualitative Data and Deep Insight

- Explanation: Interviews allow for direct, face-to-face (or voice-to-voice) interaction between the researcher and the user. This enables the interviewer to ask follow-up questions, explore deeper into interesting or unclear responses, and explore the user's reasoning, emotions, and motivations in their own words.
- Example: If a user says, "I find the checkout process confusing," an interviewer can immediately ask, "Can you tell me which specific part was confusing? Was it the payment options, the delivery form, or something else?" This leads to detailed, contextual insights that a predefined questionnaire might miss.

Disadvantage

Time-Consuming and Resource-Intensive

- Explanation: Conducting interviews requires significantly more time and effort than distributing questionnaires. Each interview must be scheduled, conducted (often lasting 30-60 minutes), and then transcribed and analyzed. This limits the number of participants you can reach within a given timeframe and budget compared to a questionnaire, which can be distributed to hundreds or thousands of people simultaneously with minimal additional cost per participant.
- Example: While you might be able to conduct 10-15 in-depth interviews in a week, you could distribute an online questionnaire that gathers data from 200+ users in the same period.

Additional questions (on your own, i.e. questions below will not be discussed during tutorial) Examine a(n) interface/computer application/website that you are UNFAMILIAR with.

(i) Which type of user(s) does it cater for?

(you only need to state whether the interface/computer application/website is designed for novice or expert user or both)

(ii) How do you know that this is the case?

(iii) How does it make you feel as a novice user of that particular system?

(iv) How could it be improved?

Examine a(n) interface/computer application/website that you are FAMILIAR with.

(i) Which type of user(s) does it cater for?

(you only need to state whether the interface/computer application/website is designed for novice or expert user or both)

(ii) How do you know that this is the case?

(iii) How does it make you feel as a user of that particular system?

(iv) Do you have any suggestion for improvement?

HCI, T3, The User's Physical Capabilities BMIT2203

Question 1

(a) Briefly explain the meaning of **cognition**.

Cognition is the process by which we interpret information in our minds. It involves taking in information from our senses (eyes, ears, touch), interpreting that information, planning an action based on the interpretation, and then executing the action. In the context of HCI, it is the set of mental processes that allow users to interact with technology.

b) Some interfaces/displays are hard to interpret (e.g. it is confusing or difficult to comprehend). Find 2 examples of such interfaces and bring it to the classroom. The 2 examples must be of websites/software developed by Malaysian companies only. Point out which part(s) of the interfaces/displays that are hard to interpret/confusing/difficult to comprehend. Explain your answer. Recommend how the problems can be solved.

The screenshot displays the PTPN (Permodalan Nasional Berhad) website. The interface is cluttered with numerous icons and text blocks. The top navigation bar includes icons for 'PERKHIDMATAN PERKHIDMATAN', 'PERKHIDMATAN PERKHIDMATAN', 'PERKHIDMATAN PERKHIDMATAN', 'PERKHIDMATAN PERKHIDMATAN', 'PERKHIDMATAN PERKHIDMATAN', 'PERKHIDMATAN PERKHIDMATAN', and 'PERKHIDMATAN PERKHIDMATAN'. The main content area is divided into several sections: 'PERKHIDMATAN' with circular icons for 'e-les', 'UJRAH', 'e-SSPN', 'POHON', 'e-CARE', and 'KAJIAN'; 'INFO TERKINI' with news articles; 'MEDIA' with a video player; 'BERSAMA KAMI' with social media links; and 'PENAFIAN & STATISTIK' with a QR code and statistics. The interface is visually overwhelming due to the high density of icons and text, making it difficult for users to find specific information.

PERKHIDMATAN

- e-les
- UJRAH
- e-SSPN
- POHON
- e-CARE
- KAJIAN

INFO TERKINI

Vaganza! Siri Kedua Tahun 2015. [Selengkapnya](#)

TERKINI Senarai Pemegang Cabutan Wow! SSPN+ Plus Siri Pertama Tahun 2015. [Selengkapnya](#)

TERKINI Untuk makluman, terdapat gangguan capaian ke atas permohonan online. Bagi pelajar-pelajar yang tidak berjaya menghantar permohonan bagi tarikh tutup permohonan 30/06/2015, proses permohonan akan di lakukan pada 15/06/2015. Semakan tarikh tutup permohonan mengikut IPT boleh disemak di alamat berikut: [http://www.usnib.gov.my/portal/berita-tutup-permohonan](#)

TERKINI Untuk memastikan transaksi bayaran balik / simpanan dimasukkan dalam penyisiran pada bulan semasa, transaksi melalui agensi/ahad katika hendaklah dibuat: [sawlat-lowabnya](#) pada 27

MEDIA

Syabas diucapkan kepada...

BERSAMA KAMI

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[Ri: Transaksi Perkhidmatan Online PTPN](#)

[Parafan | Dasar Privasi & Keselamatan](#)

[Peta Laman | Beritua Portal](#)

Sesuai dipapar menggunakan resolusi 1024 x 768,

platform IE11, Mozilla Firefox

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PTPTN Website: Analysis and Recommendations

Problem 1: Cluttered Layout and Poor Visual Hierarchy

- **Issue:** The interface is crowded with numerous icons, links, and sections placed too close together. This creates information overload and violates design principles for attention. The lack of a clear visual hierarchy means users cannot easily determine what is important, forcing them to scan the entire page inefficiently.
- **Solution:** Simplify the layout by using grouping and spacing (Gestalt Law of Proximity) to chunk related content. Prioritize key information by making it more prominent through size, colour, and placement to guide the user's attention effectively.

Problem 2: Inconsistent and Outdated Visual Design

- **Issue:** The design uses inconsistent icon styles and clashing colours, which makes the interface feel unprofessional and difficult to parse. This violates the Gestalt Law of Similarity, as dissimilar elements fail to form coherent groups, increasing cognitive load.
- **Solution:** Modernize the interface with a cohesive colour scheme and a consistent set of icons. Ensure that all similar functions have the same visual style. To reduce clutter, use progressive disclosure—for example, by hiding secondary links in a dropdown menu that appears only on hover.

Problem 3: Poor Accessibility and Responsiveness

- **Issue:** The website uses small text and has low colour contrast between the text and background, making it difficult to read. Furthermore, the layout is not responsive, creating a poor experience for mobile users and those with visual impairments.
- **Solution:** Improve perception and readability by using larger, legible fonts and ensuring high contrast ratios. Implement a responsive design that adapts seamlessly to different screen sizes, adhering to guidelines for inclusive design.

Question 2

(a) Visit the link below and then carry out the following activities:

<http://architectingusability.com/2011/05/26/using-the-gestalt-laws-of-perception-in-ui-design/>

Prepare brief notes and examples for the following 2 laws

- Law of proximity
- Law of similarity (perception)

[chap4 - slide 25]

- Law of Proximity:
 - Definition: Objects that are close to each other in space or time are perceived as being related or belonging to the same group.
 - UI Example: In a navigation menu, placing menu items close together within a distinct section (like "Account Settings") groups them. If you want items to be seen as separate, ensure there is enough space between them.
 - As seen in PDF: The comparison of the well-designed form (labels close to fields) vs. the poorly designed one (labels far from fields).
- Law of Similarity:
 - Definition: Elements that share similar visual characteristics (like color, shape, or size) are perceived as belonging to the same group or having the same function.
 - UI Example: Using the same icon style and color for all "edit" buttons across an application. All clickable links are blue and underlined.
 - As seen in PDF: The file explorer where all Word documents share the same blue 'W' icon, allowing users to instantly group them visually.

(b) Comment the 2 figures below in terms of its *figure(foreground)* and *ground(background)*. (hints: using your favourite browser search for the subject “**FIGURE and GROUND**” for more information).



Principle: The Figure-Ground law states that our mind separates the visual field into the foreground (the figure, which we focus on) and the background (the ground, which recedes). Sometimes, the relationship is ambiguous and can be interpreted in two ways.

For the left image, when we focus on the white area, the foreground shows a white vase.

While when we focus on the black area, or also the background we can see there is 2 human faces facing each other.

For the right image, we can see 2 white human faces facing each other when focusing on the black outline, while focusing on the white area shows a white vase.

(c) “For UI design, the *ground(background)* of the UI should be designed in such a way that it supports the *figure(foreground)*.” Do you agree with this statement? Explain your answer.

Yes, I strongly agree with this statement.

- Explanation: The foreground (e.g., text, buttons, icons) contains the primary interactive content. If the background is too busy, has low contrast, or competes for attention, it undermines the legibility and usability of the foreground elements.
- The guidelines for perception emphasize "sufficient contrast between the foreground and background" and avoiding a "jumbled effect" with too many fonts. A supportive background ensures text is legible, icons are distinguishable, and the user's attention is directed to the correct elements, reducing cognitive load.

Question 3

(a) Identify the 3 common applications of sounds. State which 2 are the most commonly used in HCI. Give 2 examples (own examples) and elaborate your answer.

[chap3 - slide20]

- 3 Common application
 - As Means to LOCATE Things
 - As Means to provide FEEDBACK
 - As means to attract ATTENTION

The two most commonly used in HCI are Feedback and Attracting Attention.

- Example for Feedback: The "click" sound when pressing a smartphone's virtual keyboard confirms that the keypress has been registered.
- Example for Attracting Attention: A loud, distinctive alarm sound from a factory control system alerts an operator to a critical error that requires immediate action.

(b) Give 2 examples of systems (own examples) that should use sound for FEEDBACK and/or ATTRACTING attention but unfortunately currently it is not using it. Justify your answer.

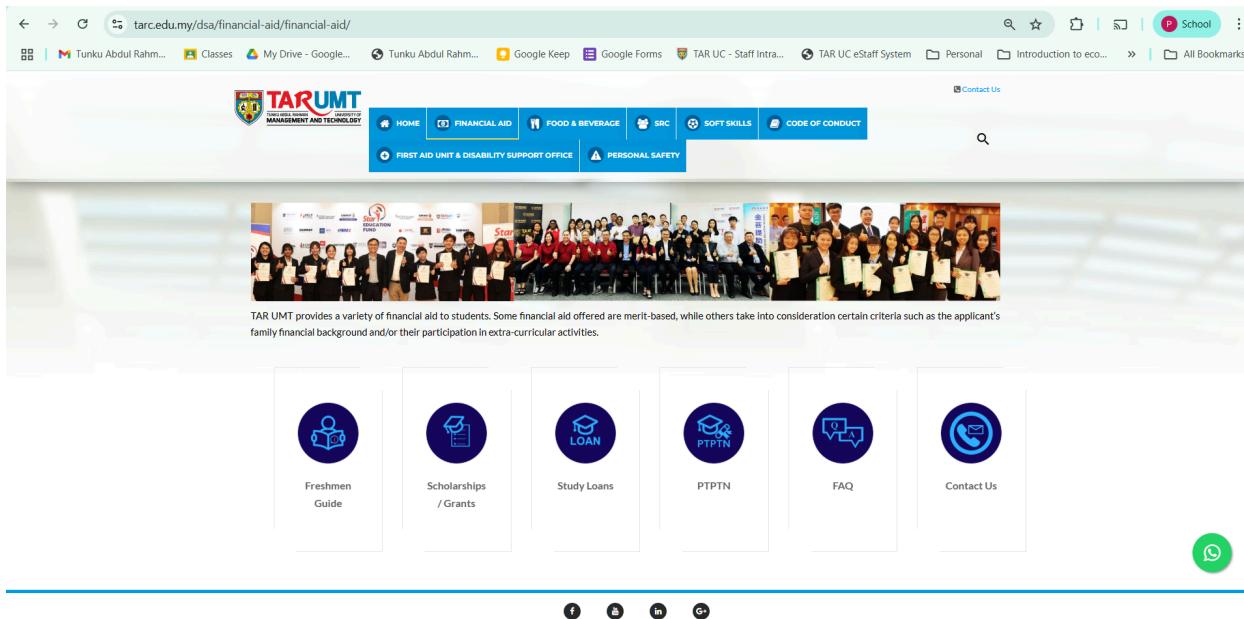
Feedback

- Example 1: Silent Electric Kettle
 - Justification: Many modern electric kettles have no audible signal when they finish boiling. The user must constantly check the kettle or stay within visual range. An auditory signal (like a whistle or beep) would be highly effective as **feedback** that the task is complete, especially if the user is in another part of the kitchen.
- Example 2: Desktop "Save" Operation
 - Justification: In many word processors or graphic design software, when you click "Save," there is often no confirmation beyond a small icon change or progress bar. In a multi-monitor setup or when the user is distracted, a subtle, non-intrusive sound as **feedback** (e.g., a soft "swoosh") would provide immediate, non-visual confirmation that the file has been saved successfully.

ATTRACTING ATTENTION

- Example 3: Critical System Update Alert
 - Justification: Operating systems often rely on passive visual indicators (like a small icon in the system tray) to notify users of critical security updates that require a restart. A user who is fully immersed in a task, such as writing a document or coding with a full-screen application, can completely miss these visual cues for hours or days, leaving their system vulnerable. This is a scenario where an immediate response is needed for security. A distinctive, non-ignorable sound (similar to a low-battery warning) would be highly effective in attracting the user's attention away from their current task to address the critical alert, especially since the visual system is already overburdened.

Past Year Question (Group 4)



Question 2 (Continued)

- c) Point out an example of the application of Fitts' Law based on the Figure 1 above. (5 marks)

Question 2

You have been appointed to design a food ordering mobile app to tailor for university canteens. It allows university students to browse menus, customise orders, and make payments directly from their smartphones. The app supports real-time order tracking to minimise waiting time, provides updates on preparation and estimated pickup times.

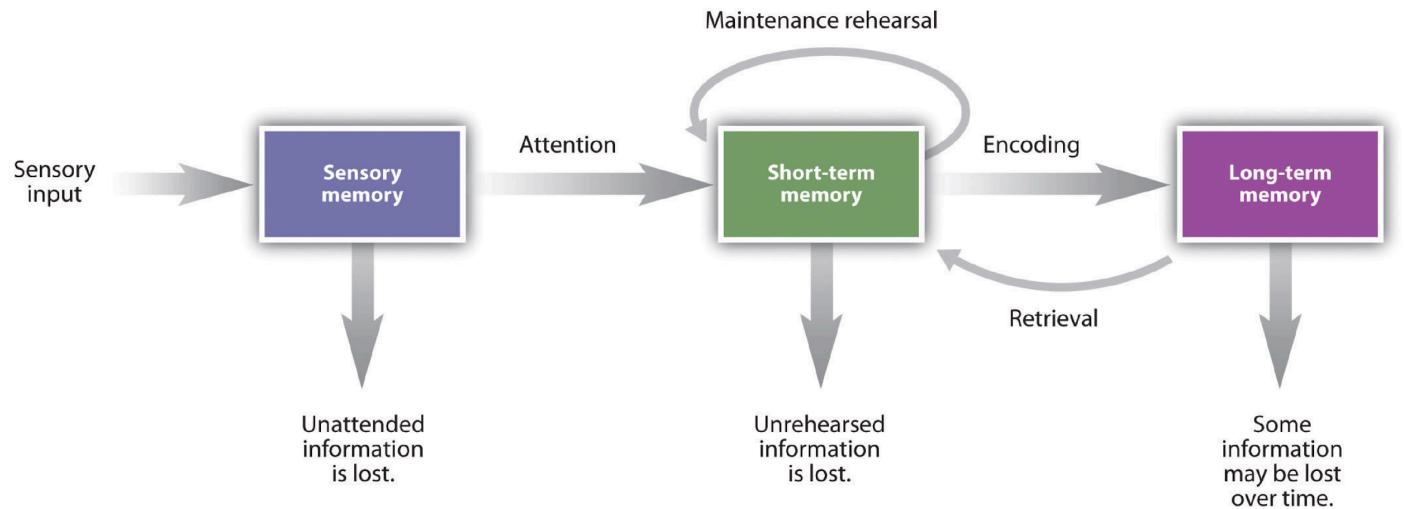
Answer the following questions based on the scenario above.

- a) Identify **THREE (3)** common usages of sound and discuss **ONE (1)** example of how to apply sound in a food ordering app. (5 marks)

HCI, T4, The User's Mental Capabilities BMIT2203

Question 1 (Group 5)

(a) Draw a diagram to illustrate the 3 types of human memory. Explain your answer.



Explanation:

Information from our senses enters Sensory Memory (SM).

If we pay attention to it, it moves to Short-Term Memory (STM), which has a limited capacity (7 ± 2 items) and duration.

Through rehearsal or encoding, information can be transferred to Long-Term Memory (LTM), which has a virtually unlimited capacity.

When needed, we retrieve information from LTM back into STM to use it.

(b) Complete the table below to highlight the differences between STM and LTM.

	STM	LTM
Capacity	Limited (7 ± 2 chunks)	Unlimited
Access	fast	slow
Decay	Fast (unrehearsed info is lost)	Slow (some info lost over time)

(c) According to Miller (1956), STM can hold between 5 to 9 chunks or pieces of information for a very short period of time. Based on Miller's studies on the capacity of STM, comment on whether it is necessary for a designer to limit the number of items displayed in a menu to a maximum of 9? Give reasons for your answer.

No, it is not strictly necessary to limit all menus to a maximum of 9 items.

- Reason: Miller's law refers to the recall of arbitrary, unconnected chunks of information. Modern GUI menus support recognition over recall. Users don't have to remember all the items; they simply scan the list and recognize the one they need. Furthermore, menus can be chunked into logical categories (e.g., File, Edit, View), which helps users process more than 9 options effectively. Forcing a limit of 9 could lead to illogical grouping or hidden menus, making the interface harder to navigate.

Question 2 (Group 1)

(a) Give 2 examples (own examples) of interfaces/displays where you are expected to remember more than what is reasonable. Explain your answer.

- Example 1: Multi-Page Forms without Summary
 - Explanation: A tax filing website that requires you to fill in financial details across 5 different pages. On the final submission page, it asks you to confirm the accuracy of all the data without showing a summary. This forces you to recall specific numbers from your LTM from previous pages, which is unreasonable and error-prone.
- Example 2: Command-Line Interfaces (CLI) vs. Graphical User Interfaces (GUI)
 - Explanation: Using a CLI (like Windows Command Prompt or Linux Terminal) often requires users to recall exact commands and syntax from their LTM. In contrast, a GUI provides visual menus and icons that the user can recognize, which is a much easier cognitive task. Requiring recall of complex commands places an unreasonable burden on the user's memory.

(b) Give 2 recommendations on how the interfaces/displays you identified in Question 2 (a) should be designed to avoid forcing the user to remember more than what is reasonable. Discuss whether these recommendations must be followed all the time.

1. Promote Recognition over Recall: Always provide visible options, cues, and context. Use menus, icons, and autocomplete features so users don't have to remember everything.
2. Use Chunking and Grouping: Break down complex information into smaller, logical groups. For example, displaying a long credit card number as 1234 5678 9012 3456 instead of 1234567890123456 makes it easier to perceive and remember briefly.

Question 3 (Group 2)

"In HCI, mental model refers to the user's current understanding of how "something" works (e.g how a system works). Very often the user's mental model does not match with what the system actually does".

Explain the above statement by using a suitable example (the example provided must be based on your own experience).

Explain the statement about mental models using a suitable example.

Statement: "Very often the user's mental model does not match with what the system actually does".

- Explanation: A user's mental model is their internal understanding of how a system works, based on past experience and intuition. The system's actual internal processes (the designer's model) are often different. When these two models don't align, the user finds the system confusing and difficult to use.
- Example from my experience:
 - My Mental Model: When I "delete" an email, I think it goes to a "Trash" or "Bin" folder, similar to a physical wastebasket in my office. I expect I can recover it if I made a mistake, at least until the trash is "emptied."
 - The System's Reality (in a specific webmail client): I once used an email service where clicking "Delete" on an email in the "Inbox" would permanently delete it, bypassing the "Trash" folder entirely. The only way to move an email to "Trash" was to manually drag it there.
 - Mismatch: My mental model (delete -> trash) did not match the system's actual function (delete -> permanent destruction). This led to the accidental and irreversible loss of several important emails because the system's behavior violated my deeply held expectation.

Past year question (Group 2)

Question 2

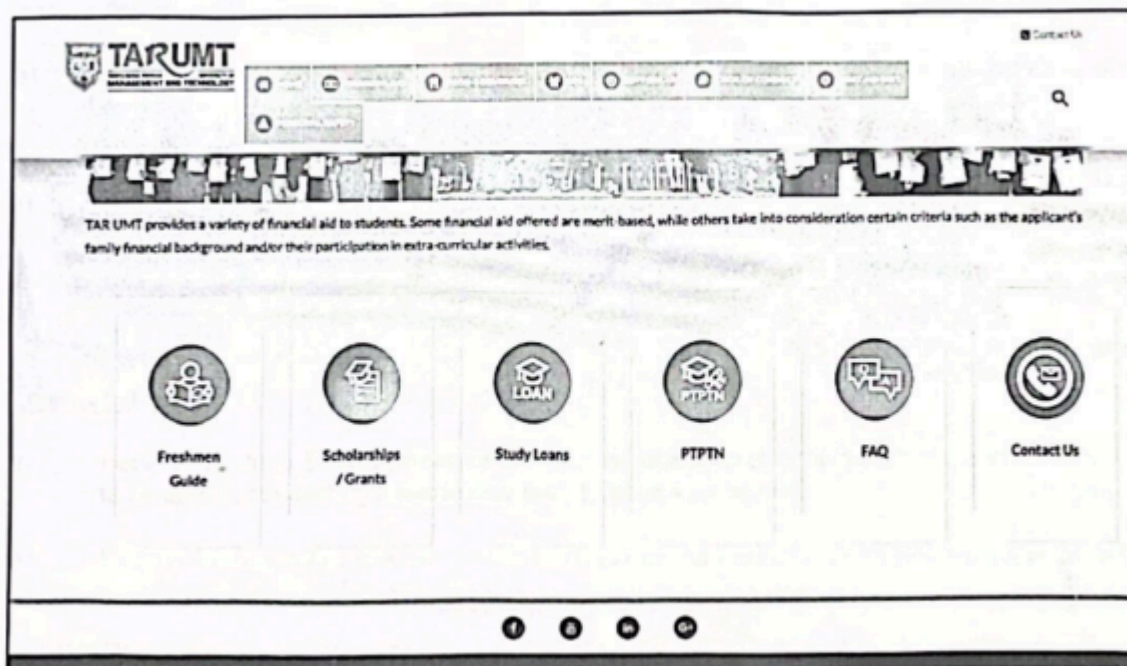


Figure 1: The Financial Aid web page from TARUMT website.
(source : <https://www.tarc.edu.my/dsa/financial-aid/financial-aid/>)

- a) Point out an example of the application of Gestalt law of "Proximity" based on the Figure 1 above. (5 marks)

Past year question

Question 2

You have been appointed to design a food ordering mobile app to tailor for university canteens. It allows university students to browse menus, customise orders, and make payments directly from their smartphones. The app supports real-time order tracking to minimise waiting time, provides updates on preparation and estimated pickup times.

Answer the following questions based on the scenario above.

- b) Explain Gestalt law of "Similarity" with an example in the food ordering app. (5 marks)
- c) Based on Miller (1956), Short Term Memory (STM) can hold 5 to 9 chunks of information for a brief period. Is it essential to design the food menu pages limited to a maximum of 9 items on each page? Justify your answer. (5 marks)
- d) Discuss with an example how the user interface design of a food ordering app facilitates recognition rather than recall. (5 marks)
- e) Provide an example of a metaphor and explain how to apply it in a food ordering app. (5 marks)

Question 2 b) (Group 3)

The Gestalt Law of Similarity states that elements that share similar visual characteristics (like color, shape, or size) are perceived to be related or belonging to the same group.

- Example in the App: All "Add to Cart" buttons for every food item can be designed with the same green color, rectangular shape, and bold white text. Even if they are separated on the screen next to different meals, users will immediately recognize that all these similarly styled green rectangles perform the same function. This creates a visual pattern that allows users to quickly find the action they want to take without reading every label.

Question 2 c) (Group 3)

No, it is not essential to limit menu pages to a maximum of 9 items.

- Justification: Miller's law refers to the recall of arbitrary, unconnected pieces of information. A menu is a classic example of an interface that supports recognition over recall. Users do not need to hold all menu items in their STM; they simply scan the list and recognize what they want. Furthermore, items can be chunked into logical categories (e.g., "Rice Dishes," "Beverages," "Desserts"), which helps users process information far beyond 9 items. Artificially limiting to 9 could result in excessive scrolling and navigation, making the app less efficient.

Question 2 d) (Group 4)**Question 2 e) (Group 4)****Past year question (Group 5)**

Question 2



Figure 1: The search result of iDiscover from TAR UMT library. (source : <https://library.tarc.edu.my>)

- b) Explain with an example of using this website could potentially overtaxing a user's short-term memory (STM) based on the Figure 1. Based on your given example, propose ONE (1) solution to help the users. (5 + 5 marks)

HCI, T5, UI Design Guidelines BMIT2203

Question 1

One of the Shneiderman's 8 golden rules of interface design is "*Strive for consistency*". (a) Explain what this golden rule means.

(b) Explain what you or your group **must do** in order to apply this guideline to your HCI assignment (i.e. your HCI assignment's prototype).

Question 2

(a) Explain in details the good things and the bad things about the *Shneiderman's 8 golden rules* of interface design.

(b) In your opinion are these UI design guidelines useful to software developers? Justify your answer.

Question 3 (*Present your answers using PowerPoint slides*)

(a) Download and read the article "*The Case Against User Interface Consistency*" by Jonathan Grudin. (read pages 1164-1167 is enough).

<https://www.microsoft.com/en-us/research/publication/case-user-interface-consistency/>

Based on the above article, mention an **example of one of the best designs that violate the guideline "*consistency*"**. Explain your answer.

(b) Give an example (**own example**) of a good UI design that violate the guideline "*consistency*". Explain your answer.

Additional

questions (on your own, i.e. questions below will not be discussed during tutorial) Question 4 (AAC5194(B) - 06/07 April/May Exam)

Suggest one situation where it is impossible to provide a consistent user interface.

Question 5 (AAC5194(A) - 06/07September Exam)

Comment on how the Shneiderman's 8 golden rules of interface design may be used in the HCI design process.

HCI, T6, Techniques To Draw The User's Attention BMIT2203 and Error Messages

Question 1 (*Present your answers using PowerPoint slides*)

- (a) Susie is a new programmer working with you in a software house. She would like to know how to use colours effectively for screen output. Give her some guidelines that might be useful to her. For each guideline, give an example (**own example**) to convince her.
- (b) You have been asked to help in the development of an online Stocks System. **Table 1** below shows some of the important information that must be published on the main page i.e. the most active stocks for a particular trading day. Your software manager would like to seek your advice on how to use proper colours to enhance the usability of the Stocks System.

Table 1: Most Active Stocks

Stock Code	Name	Volume	Changes
8200	GENETECH	2,500,000	+1.60
0028	AIR COMMUNICAT ION	1,200,000	+0.80
5303	IRIS	900,000	0.00
5088	YXL	80,000	0.00
7110	CHEE KEE	60,000	-1.50

Required:

Demonstrate how you may apply colours to the information given in Table 1 to enhance the usability of the Stocks System. Justify the choice of colours used.

Question 2 (*Present your answers using PowerPoint slides*)

- (a) Find any 5 artefacts (e.g. *signboard, signage, announcement on noticeboard, etc*) in your campus. Take picture of the artefacts and insert them into PPT slides. Analyse the color(s) used and based on what you have learned in HCI, comment whether the color(s) used is good or poor. Give reasons to support your answer.
- (b) Explain what *closure* means.
- (c) "*Closure requires feedback*". Do you agree with the above statement? Explain your answer by using one suitable example (**own example**). In addition, discuss the consequences where there is little or no feedback provided.

Question 3 (*Present your answers using PowerPoint slides*)

According to Nielsen (2001), good error messages are *Explicit, Human Readable, Polite, Precise* and *Constructive*.

- (a) Give 2 examples (**own examples**) of error messages that conform to some or all of above characteristics. State the source of your answers.
- (b) Differentiate between a mistake and a slip. Give an example (**own example**) of each. Suggest how both (mistake and slip) can be avoided.

questions (on your own, i.e. questions below will not be discussed during tutorial)

Question 4 - Discuss one important factor that a designer should take into account when designing error messages for games.

Question 5 - Describe the characteristics of good error messages.

HCI, T6, Techniques To Draw The User's Attention BMIT2203 and Error Messages

~~Question 6~~ ~~Call to action in web design~~ and in user experience (UX) in particular is a term used for elements in a web page that solicit an action from the user. The most popular manifestation of call to action in web interfaces comes in the form of clickable buttons that when clicked, perform an action (e.g. "Buy this now!") or lead to a web page with additional information (e.g. "Learn more...") that asks the user to take action. Discuss some of the techniques to draw users' attention and lead them to call to action.

Question 7 *(present your answer using PowerPoint slides)*

Colour if used appropriately can make an interface pleasant and enjoyable to look at. Common applications of colour includes:

- highlight differences between information
- draw attention
- indicate status

Give 2 examples for each of the above. You may search the Internet or your Windows OS to find the required examples. Comment on the color used.

HCI, T7, Input and Output Devices BMIT2203

Question 1

(a) Explain the following keyboard layouts:

- QWERTY
- DVORAK

(b) *Computer keyboard* layout followed *typewriter keyboard* layout. Why?

(c) Describe 2 situations in which *keyboard* might be preferred over *speech* for data entry/issuing command. Briefly explain your answer.

Question 2

(a) Describe 2 situations in which *speech* might be preferred over *keyboard* for data entry/issuing command. Briefly explain your answer.

(b) Explain why thermal printer is a better choice for ATM machines.

Question 3 (*Present your answers using PowerPoint slides*)

(a) “*If a fill-in form have both text entry fields and multiple-choice selection fields then each input type should be grouped together.*”

Do you agree with the above statement? Give reasons and an example (**own example**) to support your answer.

(b) Watch <https://www.youtube.com/watch?v=jbV5dGvJWyo>

Based on the video, list some of the common sense things that you should do to avoid neck, shoulder, arm and leg pains as well as eyestrains when using a computer. In other words, give advice on how to avoid health problems when using a computer.

(c) Find a similar video in youtube and **share** it with everyone in the class. Video title:

Additional

questions (on your own, i.e. questions below will not be discussed during tutorial) A PC usually equipped with a QWERTY keyboard, a mouse and a VDU. Would these be satisfactory interactional devices? What other input and/or output devices might be needed and why? (You may use a scenario or make an assumption to support your answer)

Discuss the types of input and output devices that you would use for the following system. For each system, indicate why the conventional keyboard, mouse and computer screen may be less suitable. (You should do some research to answer this question).

- (a) Global Positioning System (GPS)
- (b) Automated Teller Machine (ATM)
- (c) Tourist information system

HCI, T8, Menus & Fill-in Form BMIT2203

Question 1

A common form of interaction is through the use of menus. Discuss this form of interaction in terms of:

- (a) General advantages and disadvantages of menus (give 2 advantages and 2 disadvantages of menus)
- (b) List all the ways of ordering/organising menu items. For each way give an example (**own example**) to support your answer.
- (c) Do some research to find out other possible ways/choices of ordering menu items. List out those ways/choices not covered in the lecture. Explain your answer.

Question 2 (*Present your answers using PowerPoint slides*)

- (a) Linda is a new programmer working with you in a software development company. She requested assistance from you to give her some advice on how to design an effective form. Write 5 design guidelines for fill-in form that you think might help her. Explain your answer.
- (b) Find a fill-in form (onscreen or paper fill-in form) which you think is well designed and another one which you think is poorly designed. Share with the class both forms during tutorial. Explain your answer.

Question 3

Based on what you have learned in HCI, design a new *examination attendance slip* to replace the existing one. The examination attendance slip must contain the following fields:

Seat no, Index no (in words), Index no (in figures), Paper examined (as printed on the exam paper), Date of examination, Time of examination, Signature.

Show how it would appear on paper and briefly explain the rationale of your design.

Additional

questions (on your own, i.e. questions below will not be discussed during tutorial) “Mobile devices required different menu design”
Comment on the above statement. Discuss some of the interaction styles and highlight some design considerations of menu design on mobile devices.

HCI, T9, Questionnaires BMIT2203

Question 1

- (a) In the context of HCI, explain the meaning of evaluation. Give 3 reasons why it is important to evaluate systems.
- (b) What are the methods that a designer can use to evaluate the usability of a product/system/software?

Question 2

- (a) In HCI, questionnaires have long been used to evaluate user interfaces. Discuss 2 advantages and 2 disadvantages of using questionnaires as a technique of evaluation.
- (b) One of the issues of using questionnaires is poor returned rates. Suggest 2 solutions to overcome this problem.

Question 3

- (a) Analyse the questionnaire given below and then redesign the questionnaire to improve it. You must show at least 4 improvements in your answer. Give reasons for your answer. *(Note: Do NOT change the last question i.e. let it be an open question)*

- (b) Suggest a cost effective way to evaluate the usability of the Google Classroom. Justify your answer. Include also in your answer the steps to be followed to carry out the evaluation.

HCI, T10, Expert Reviews and Observing Users BMIT2203

Question 1

Experts (UI/HCI Specialists) can be involved in the design of user interfaces. (a) Explain how they can help developers to produce better user interfaces. (b) Evaluate the advantages and disadvantages of involving experts?

Question 2

Many large software development companies have usability laboratories to test their software.

- (a) Discuss the pros and cons of testing software in a usability laboratory.
- (b) How many tester(s) is/are needed for the first test and subsequent tests. Give reason for each of your answers.

Question 3

- (a) Compare and contrast between *field studies* and *usability testing in a usability laboratory*.
- (b) For field studies, what do you think are the right things to do in order to minimise the *Hawthorne/Observer Effect*?
- (c) Fill-in the empty cells for the table below:

Evaluation and testing	Techniques
(1) with users involvement	
(2) without users involvement	

HCI, T11, Usability Measurements, Ergonomics BMIT2203

Question 1 (Present your answers using PowerPoint slides)

- (a) Explain how usability measurements can be beneficial to both developers and customers. (b) What are the ways of measuring software usability?
- (c) Suggest suitable way(s) to measure the usability of your HCI assignment's prototype. Justify your choice(s).

Question 2 (Present your answers using PowerPoint slides)

- (a) What are the benefits of making the working environment ergonomically correct?
- (b) Describe the negative impact to workers as a result of a poorly designed working environment while at the same time they have to use a computer over a long period of time.
- (c) Using your favourite browser search for "Google Office". Create about 10-20 slides to illustrate the working environments in Google.Com. Include also some photos of working environments in 3 other companies for comparison purposes (one of them must be own your place of study). Explain your answer.

Question 3 (Present your answers using PowerPoint slides)

- (a) The followings are important factors that need to be taken into account in order to create an ideal/wholesome working environment.
- Proper desks and chairs
 - Adequate lighting
 - Low noise level
 - Temperature (15°C to 22°C)

Give reason(s) why each of the above factor is important to the office workers. (b) Give one example

(own example) of an incorrect sitting posture. Explain your answer.

- (c) Give one example (own example) of a correct sitting posture. Explain your answer. *Note: For Q3(b) and Q3(c), search the Internet/textbook for the required answers.*

Additional

questions (on your own, i.e. questions below will not be discussed during tutorial) *Question 1*

(a) Do you agree that the introduction of computers can cause some of the followings? - Stress and anxiety - Fear of unemployment - Loss of responsibility - Inadequacy - Lack of privacy - Alienation - System failures - Deterioration of self image

(b) Discuss how the above problems can be mitigated.

Question 2

What is RSI and why is it an increasing problem for the computer users?

Question 3

You have been requested to produce a leaflet to show what a good typing posture looks like. Sketch a first version of this leaflet with textual explanation indicating what a good posture should look like

- End -